AquaNexis — Round 2 Progress Report

Track 2: Water Purifier | Institution: UPES Dehradun | Participation: Solo Prototype (Software-Only) | Date: 4 Nov 2026 | atomQUEST 2026

1. Project & Participant Details

Field	Value
Project Name	AquaNexis — Predictive Water Intelligence System (Software-Only Prototype)
Track & Institution	Track 2: Water Purifier | UPES Dehradun | Solo Prototype
Primary Exploration	Prognostics of filter failure using data analytics (FSI, FHI, RUL, Failure Risk, Explainability)
Secondary Exploration	Input water quality sensing beyond TDS (TDS, EC, pH, Turbidity, Temperature, Flow, Inlet/Outlet Pressure)

2. Problem & Proposed Solution

Problem: Conventional water purifiers rely on fixed calendar timers or basic TDS checks, resulting in either premature filter replacement or unexpected membrane clogging and unmonitored water quality degradation.

Solution: AquaNexis is a software-only predictive intelligence platform that ingests 13-sensor purifier telemetry, validates sensor quality, computes a normalized 7-component Filter Stress Index (FSI) and 5-component Filter Health Index (FHI), estimates Remaining Useful Life (RUL) via an explicitly disclosed deterministic fallback (**MODEL_MISSING**), classifies failure risk with feature-level evidence, and drives stateful alerts and guided maintenance workflows. All values carry explicit provenance (**SIMULATED / CALCULATED / MODEL_PREDICTED / MEASURED**).

3. Completed Software Modules

Module	Status	Verified Implementation Details
Telemetry & Quality Gate	Complete	FastAPI ingestion, JWT/API-key auth, range/rate/cross-sensor validation (GOOD/SUSPECT/BAD)
Water Quality (Beyond TDS)	Complete	AquaScore (0-100), TDS, EC, pH, Turbidity, Temperature sub-scores, 1h trends, anomalies
Filter Stress Index (FSI)	Complete	7-component normalized weights (sum=1.00); excludes missing sensors from weight denominator
Filter Health Index (FHI)	Complete	5-component composite health score (0-100) + maintenance bands (NORMAL/MONITOR/REPLACE)
RUL & Risk Prognostics	Complete	Deterministic fallback labeled MODEL_MISSING; withheld (N/A, LOW confidence) on SENSOR_FAULT
Explainability Engine	Complete	Top 5 feature contributors (delta-P, flow deviation, volume, age, turbidity) + counterfactuals
Alert State Machine	Complete	DETECTED -> OPEN -> ACKNOWLEDGED / SNOOZED / RESOLVED / AUTO_CLEARED + evidence drawer
Maintenance Lifecycle	Complete	11-category filter catalog, compatibility engine, replacement/calibration/verification workflows
WebSocket & React UI	Complete	React 18 + TypeScript dashboard (localhost:3000), live WebSocket (/ws/realtime/{id}), DEMO_MODE switcher

4. Original Sourcing Risks, Scope Reset & Budget (Estimate vs Actual)

Item / Risk	Stage 1 Estimate	Actual Spend / Status	Resolution & Honest Disclosure
ESP32 + 13 Sensors + PCB	Rs. 5,000 - 15,000	Rs. 0 (Not purchased)	Descoped in G6 to software-only prototype; firmware/esp32/ archived as reference
Physical Calibration & Rig	Bench loop planned	Rs. 0 (Simulated)	Replaced by deterministic 8-profile Python simulator (backend/simulator.py)
TimescaleDB Extension	Local PG16 + Timescale	Rs. 0 (DEMO_MODE)	Local Windows PG16 lacked timescaledb DLL; isolated DEMO_MODE runs full analytics in-memory

5. Revised Performance Targets & Verification Status

Original Target	Revised Software Target	Verification Status & Disclosure
>=4 contextual signals beyond TDS	13 simulated sensor channels	Verified: TDS, EC, pH, Turbidity, Temp, Flow, Inlet/Outlet Pressure, Tank, Leak, Current, Power, UV
RUL MAE <= 7 days	Deterministic RUL fallback + CI	Disclosed: No trained .pkl ML model yet; explicitly labeled MODEL_MISSING (Deterministic Fallback v1.0)
Dashboard update <= 5 seconds	Live WebSocket + 5s staleTime	Verified: Real-time WebSocket broadcast (/ws/realtime/{id}) + stale banner at >30s
Automatic safe shutdown	Software safety alert + audit log	Disclosed: Software alert (leak_detected) & safety log only; NO physical solenoid valve or relay exists
Bad sensor fault resilience	Quality gate + confidence guard	Verified: SENSOR_FAULT flags Quality=BAD, excludes missing sensors, and withholds RUL (Confidence=LOW)

6. End-to-End Scenario Demonstration Evidence (Verified in DEMO_MODE)

Scenario	Telemetry & Quality	Analytics Response (AquaScore / FSI / FHI / RUL / Risk)	Alert & Maintenance Action
NORMAL	Quality=GOOD, SIMULATED (dP=30 kPa, Flow=2.0 L/m)	AquaScore=100, FSI=20.0, FHI=83%, RUL=149.4d (MODEL_MISSING), Risk=LOW (13.4%)	0 active alerts; routine monitoring
FILTER_STRESS	Quality=GOOD, SIMULATED (dP=280 kPa, Flow=0.85 L/m)	AquaScore=93, FSI=67.5, FHI=40%, RUL=72.0d (MODEL_MISSING), Risk=MEDIUM (49.3%), Band=REPLACE_RECOMMENDED	Alert filter_stress_high (OPEN -> ACKNOWLEDGED); Stage 1 Sediment replacement recommended
LEAK	Quality=GOOD, SIMULATED (leak_detected=true)	AquaScore=100, FSI=22.3, FHI=83%, Risk=LOW	Critical manual-only alert leak_detected + SAFETY_SHUTDOWN_RECOMMENDED software audit log
SENSOR_FAULT	Quality=BAD, SIMULATED (Turb>150 NTU, flow/P_out=null)	AquaScore=59, FSI=0 (missing excluded), RUL=None (N/A withheld), Confidence=LOW	Alert sensor_fault_detected; prevents false confident RUL prediction

7. Automated Verification & Round 3 Roadmap

Automated Verification: Backend unit tests: **133/133 PASS** (pytest: config, security, Phase D analytics, alert state machine, auto-clear, notifications, maintenance). Frontend checks: **npm run typecheck PASS** (0 errors), **npm test PASS (14/14)**, **npm run build PASS**.

Round 3 Plan: (1) Train Random Forest / XGBoost RUL regression models on labeled run-to-failure membrane datasets to replace the deterministic fallback; (2) Deploy containerized PostgreSQL + TimescaleDB stack for multi-device fleet persistence; (3) Validate external HTTP/WebSocket telemetry ingestion with physical RO test-bench data.

Declaration: This report accurately reflects the software-only prototype. All hardware work in firmware/esp32/ is archived out of scope. All demo telemetry is simulated (backend/simulator.py), RUL uses a deterministic fallback (MODEL_MISSING), and safety actions are software alerts/logs only.